



Key science skills

The skills assessed across every Unit 3 & 4 question — from framing an aim to communicating a conclusion. Master these and you can answer the “skills” questions the content alone won’t cover.

Digital Vector EnviroSci · single-page revision summary · VCAA Study Design 2022–2026

◆ The skills examined in every question

The key science skills are the verbs of the course. They sit underneath every key-knowledge dot point in Units 3 & 4 and are fully examinable in the end-of-year exam — a question can test the skill itself, not just the content.

How to read a skills question

A content question asks *what* you know; a skills question asks you to *do* something — frame a hypothesis, identify a variable, judge validity, evaluate a method, or draw an evidence-based conclusion. The command word tells you which skill is being tested.

1 Develop aims, questions, hypotheses & predictions

Aim

A statement of **what the investigation sets out to find** — usually the effect of the independent variable on the dependent variable. *“To investigate the effect of canopy cover on soil moisture.”*

Research question

A focused, testable question the investigation answers. *“Does increasing canopy cover increase soil moisture?”*

Hypothesis

A **tentative, testable prediction of the relationship** between variables, with a direction and reasoning. *“If canopy cover increases, soil moisture will increase, because shade reduces evaporation.”*

Prediction

The **specific, expected result** if the hypothesis holds — often an “if...then...” tied to the method.

OPERATIONALISE

Variables must be measurable. “Plant health” is not — “chlorophyll content (SPAD units)” or “% leaf area healthy” is.

2 Plan and conduct investigations

Strong planning is about controlling variables and producing data you can trust.

Variable	What it is	Example (heat-island study)
Independent (IV)	The one factor you deliberately change.	Surface type (grass / asphalt / shade)
Dependent (DV)	What you measure in response.	Surface temperature (°C)
Controlled	Everything kept constant so the test is fair.	Time of day, weather, instrument, height

Controlled experiment vs control group

A **controlled variable** is held constant. A **control group** is a comparison group with no treatment. They are not the same thing — a favourite exam trap.

Repetition & sample size

More repeats and larger samples reduce the impact of random error and make results more **reproducible and reliable** — they do not, by themselves, fix a biased method.

3 Comply with safety and ethics

Safety — hazard vs risk

A **hazard** is something with the potential to cause harm; **risk** is the chance it actually does. Use the **hierarchy of controls** (eliminate → substitute → isolate → engineering → administrative → PPE), strongest first.

Ethics — fieldwork

Minimise disturbance, obtain permits, respect Traditional Owners and private land, avoid collecting protected species, and leave sites as found. Animal/human involvement triggers extra ethical guidelines.

EXAM-READY

Name a specific hazard and a matched control — “slippery creek bank → wear sturdy footwear and work in pairs” beats “be careful”.

4 Generate, collate and record data

Term	Meaning
Qualitative	Descriptive / categorical data (species present, colour, health rating).
Quantitative	Numerical, measured data (temperature, pH, count, area).
Primary	Data you generate first-hand in your own investigation.
Secondary	Data generated by others (databases, papers, agency records).

RECORD AS YOU GO

Raw data belongs in the logbook with units, date and conditions — this is what authenticates the work.

5 Analyse and evaluate data and methods

The high-mark skill cluster — turning numbers into judgements.

Accuracy vs precision

Accuracy = closeness to the true value (free of systematic error). **Precision** = closeness of repeats to each other (free of random error). A set can be precise but inaccurate.

Validity & reliability

Valid = measures what it intends to, variables controlled. **Reliable / reproducible** = consistent on repetition. Validity is about *method design*; reliability is about *consistency*.

Error

Random error scatters readings (reduce by repeating & averaging). **Systematic** error shifts them all one way (fix by calibrating / correcting the method).

Outliers & uncertainty

Identify outliers and decide — with justification — whether to retain or exclude them. Report measurement uncertainty rather than implying false exactness.

6 Construct evidence-based arguments & conclusions

What a conclusion must do

Directly answer the **aim/hypothesis**, be **supported by the data** you generated, state whether the hypothesis was supported or refuted, and **stay inside the evidence** — no claims your data can't back. Acknowledge limitations and their effect on confidence.

7 Analyse, evaluate and communicate scientific ideas

Conventions

Use correct scientific terminology, symbols, units (SI) and significant figures; choose the right representation (table, line graph for trends, bar graph for categories); reference sources; and communicate succinctly. The Outcome 3 deliverable is a **scientific poster** — see the Scientific Investigation page.

◇ Command-word decoder

Verb	What the marker wants
Identify / State	Name it — no explanation.
Describe	Features / what happens, in order.
Explain	The how / why — cause and effect.
Distinguish / Compare	Make the difference (and similarity) explicit.

Analyse	Break into parts and show relationships.
Evaluate / Justify	Weigh strengths & limitations, then give a reasoned verdict.

★ Self-check

Click to reveal a weak vs strong response.

Distinguish between accuracy and precision. (2 marks)

Weak: Accuracy is being correct and precision is being precise.

Strong: Accuracy is how close a measurement is to the true value (small systematic error); precision is how close repeated measurements are to one another (small random error). A set of readings can be precise yet inaccurate if a systematic error shifts them all.

A student measured soil pH once at each site. Evaluate the reliability of this method. (3 marks)

Weak: It's not reliable because they only did it once, so they should do it more.

Strong: Reliability is low: a single reading per site cannot reveal random variation, so the result is not reproducible. Taking several readings per site and averaging would reduce random error and allow consistency to be assessed, improving reliability — though it would not, alone, correct any systematic bias in the probe.



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